AOU
الجامعة العربية المفتوحة
Arab Open University
Faculty of Computer Studies

T216B
Cisco Networking (CCCNA) part B

Final Examination
Spring Semester 2024/2025

Date: 03/June/2025

Number of Exam Pages: 5 (including this cover sheet)	**Time Allowed:** 3h

Instructions:

- Total Marks: **50**
- This exam consists of **3** parts.
- **ALL questions** must be answered in the External Answer booklet.
- Be sure you write your **name and ID** on the External Answer booklet.
- **Calculators** are allowed.

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Part 1: Multiple choices. 1 mark per question. **Total: 6 Marks**

1) What does NAT overloading use to track multiple internal hosts that use one inside global address?
   a. Port numbers
   b. Autonomous System numbers
   c. MAC addresses
   d. IP addresses
2) Why is QoS an important issue in a converged network that combines voice, video, and data communications?
   a. Legacy equipment is unable to transmit voice and video without QoS.
   b. Data communications are sensitive to jitter.
   c. Data communications must be given the first priority.
   d. Voice and video communications are more sensitive to latency.
3) Which of the following statements is a function of Hypervisor?
   a. To partition the hard drive to run virtual machines
   b. To protect the host from malware infection from the virtual machines
   c. To allocate physical system resources to virtual machines
   d. To share the antivirus software across the virtual machines
4) Which statement describes the concept of cloud computing?
   a. Separation of application from hardware
   b. Separation of operating system from hardware
   c. Separation of control plane from data plane
   d. Separation of management plane from control plane
5) A computer can access devices on the same network but cannot access devices on other networks. What is the probable cause of this problem?
   a. The computer has an invalid IP address.
   b. The computer has an incorrect subnet mask.
   c. The cable is not connected properly to the NIC.
   d. The computer has an invalid default gateway address.
6) Which ACE will permit a packet that originates from any network and is destined for a web server at 192.168.1.1?
   a. Access-list 101 permit tcp host 192.168.1.1 any eq 80
   b. Access-list 101 permit tcp host 192.168.1.1 eq 80 any
   c. Access-list 101 permit tcp any host 192.168.1.1 eq 80
   d. Access-list 101 permit tcp any eq 80 host 192.168.1.1

Part 2: Answer all the questions – 4 marks per question **Total: 20 Marks**

1) Answer the following questions:
   a) Match the security terms to their corresponding descriptions; not all the terms will be used. (2 marks)

Security terms:

• Asset • Mitigation	• Exploit • Threat	• Vulnerability

	Description	Security term
A	It's a weakness in a system, or its design, that could be exploited by a threat	
B	It's the counter-measure that reduces the likelihood or severity of a potential threat or risk.	

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C	It's a potential danger to a company's assets, data, or network functionality.	
D	It's anything of value to the organization. It includes people, equipment, resources, and data.	

b) Explain the following type of Attack: ***Reconnaissance attack.*** (2 marks)

2) Answer the following questions:
   a) There are two types of Access Control Lists (ACLs): standard ACL and extended ACL. Discuss where each of these two types should be placed in the network. (2 marks)
   b) What is an Access Control List (ACL)? (2 marks)
3) Answer the following questions:
   a) What is Network Address Translation (NAT)? (2 marks)
   b) Discuss the following two types of Network Address Translation: Static NAT and PAT. (2 marks)
4) Answer the following questions:
   a) List two benefits to using VPN. (2 marks)
   b) Explain what is meant by Remote access VPN. (2 marks)
5) Answer the following questions:
   a) Having received the following syslog message (figure A), determine:
      - MNEMONIC
      - Severity
      - Timestamp
      - Facility

      (2 marks)

*Mar 1 11:52:46: %LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0/0, changed state to up
Figure A

   b) When selecting a switch there are many business considerations that should be taken into account.

Considerations:

• Scalability • Frame Buffers	• Port density • Power	• Reliability

Match each consideration to its appropriate description; not all the considerations will be used. (2 marks)

	Description	Consideration
A	It is used to power access points, IP phones, and compact switches.	
B	The switch should provide continuous access to the network.	
C	Network switches must support the appropriate number of devices on the network.	
D	The ability of the switch to store frames is important in a network where there may be congested ports to servers or other areas of the network.	

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Part 3: Answer all the following questions **Total: 24 Marks**

1) Configure OSPF (13 marks)

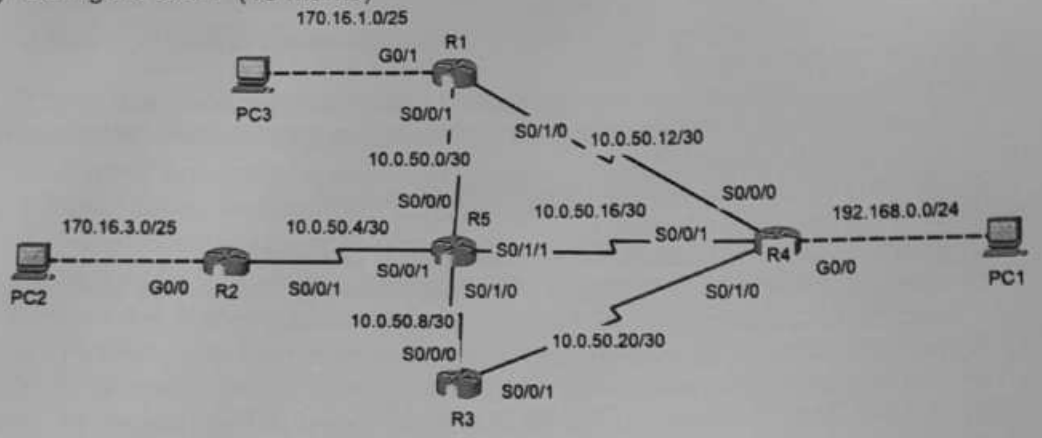


Figure 1

Scenario
OSPF was configured on all routers except R4. All the directly connected networks were advertised.

A) Determine the number of networks that are directly connected to R4 and the number of remote networks for R4? (2 marks)
B) List all the directly connected networks to R4 (the prefix (network address) and the prefix length). (2 marks)
C) Enable OSPF on router R4 (refer to figure 1) (1 mark)
Note: the process ID is 33
D) Write the command that should be used to configure the router ID 175.25.2.3 on router R4 (1 mark)
E) Advertise all the directly connected networks of router R4 except the one connected to the Internet. (Refer to figure 1) (4 marks)
F) Disable the advertisement of OSPF only on R4 interface G0/0 (1 mark)

On router R5
G) On router R5 configure a default static route via the exit interface S0/0/1 so that all traffic will be forwarded to R2 (1 mark)
Note: you are in the global configuration mode
H) On router R5 advertise the default static route to all other routers. (1 mark)
Note: you are now in the router ospf configuration mode

2) Answer the following questions (11 marks)
A) In figure 2 we have two LANs, 192.168.50.0/24 and 192.168.51.0/24 attached to R1. Router R2, the border router, is configured for Dynamic Network Address Translation (NAT) using a pool of public IPv4 addresses 209.165.200.226 through 209.165.200.240.

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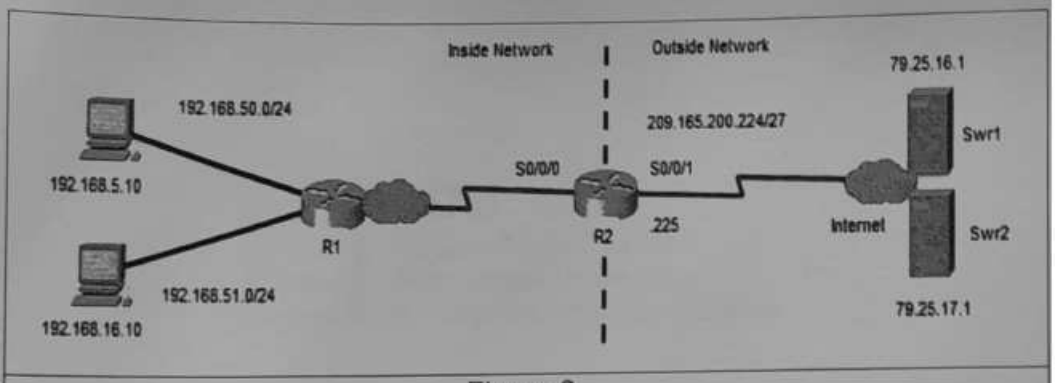


Figure 2

Complete the missing parts of the configuration: (5 marks)

```
R2(config)#ip nat pool CONF-R2 __A__ 209.165.200.240 netmask
255.255.255.224
R2(config)#access-list 1 __B__ 192.168.50.0  0.0.0.255
R2(config)#access-list 1 permit 192.168.51.0  0.0.0.255
R2(config)# ip nat inside source list 1 __C__ CONF-R2
R2(config)# interface __D__
R2(config-if)# ip nat inside
R2(config-if)# exit
R2(config)# interface __E__
R2(config-if)# ip nat outside
R2(config-if)# end
R2#
```

B) The network Administrator wants to configure a Standard ACL on router R2. Write the commands needed to configure a standard ACL to complete the following tasks: (6 marks)

- Forbid traffic from the following IP address 172.16.1.10
- Allow traffic from the following IP address 192.168.0.10
- Forbid traffic from the following network 192.168.0.0/24
- Forbid traffic from the following network 10.0.0.0/24
- Permit all other traffic
- Appy the ACL on traffic entering interface S0/0/1

End of questions

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